ModelIt!

Teacher's Guide

Genetic Mutations: When DNA Makes a Typo

How UV Radiation and Cell Division Create Errors That Change Organisms

Quick Reference

Grade Level	Grade 08L2
NGSS Standard	MS-LS3-1, MS-LS3-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: MS-LS3-1, MS-LS3-2

Develop and use a model to describe why structural changes to genes (mutations) located on chromosomes may affect proteins and may result in harmful, beneficial, or neutral effects to the structure and function of the organism. Develop and use a model to describe why asexual reproduction results in offspring with identical genetic information and sexual reproduction results in offspring with genetic variation.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a model showing how environmental and cellular factors affect mutation frequency and its consequences for organisms.
Disciplinary Core Idea	**LS3.A Inheritance of Traits** — Genes are located on chromosomes and can be altered by mutations, which may change the proteins they produce and affect organism traits.
Crosscutting Concept	**Cause and Effect** — Students identify UV radiation and cell division rate as CAUSES of mutations, and protein function changes as EFFECTS that determine organism fitness.

CAST Testing Alignment

MULTIPLE CHOICE:

A mutation changes one amino acid in a protein. The organism shows no change in appearance or survival. Which type of mutation is this most likely to be?

CONSTRUCTED RESPONSE:

Using evidence from your model, explain how UV radiation exposure and cell division rate both contribute to increased mutation frequency. Describe how these mutations can lead to changes in protein function and organism fitness.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don’t need to teach all of this — it’s here so you can answer student questions accurately.

DNA is the instruction manual for building every protein in your body. When DNA is copied during cell division, errors can occur. Environmental factors like UV radiation increase these errors. Understanding mutations is essential for genetics, medicine, cancer research, and evolutionary biology.

How Mutations Happen

DNA replication is remarkably accurate — DNA polymerase makes approximately 1 error per billion base pairs copied. However, with 3 billion base pairs per cell and millions of cell divisions per day, errors accumulate. UV radiation causes thymine dimers (chemical bonds between adjacent bases) that can lead to incorrect bases being inserted during replication. Chemical mutagens, radiation, and viral infections also increase mutation rates.

Types of Mutations

Point mutations change a single base pair. Insertions add extra bases. Deletions remove bases. Frameshift mutations (insertions or deletions) are often the most damaging because they shift the reading frame, changing every amino acid downstream. Chromosomal mutations involve large segments of DNA being duplicated, deleted, inverted, or translocated.

Effects on Proteins

Since the DNA sequence determines the amino acid sequence of a protein, mutations can change the protein's shape and function. Silent mutations don't change the amino acid (due to redundancy in the genetic code). Missense mutations change one amino acid. Nonsense mutations create a premature stop signal, truncating the protein. About 70% of point mutations are neutral, 29% are harmful, and less than 1% are beneficial.

Mutations and Cancer

Cancer occurs when mutations accumulate in genes that control cell growth (oncogenes and tumor suppressor genes). A single mutation usually isn't enough — cancer typically requires 5-10 mutations in critical genes over a lifetime. This is why cancer risk increases with age (more time for mutations to accumulate) and with UV exposure, smoking, or chemical exposure (more mutations per year).

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**
Students identify UV radiation exposure, cell division rate, mutation frequency, protein function change, and organism fitness as system components.

E **Establish Relationships**
Students determine that UV radiation and cell division rate both increase mutation frequency, mutations can change protein function, and protein changes affect organism fitness positively or negatively.

V **Visualize & Model**
Students build the mutation model showing how environmental and cellular inputs drive genetic changes that affect organisms.

E **Evaluate Outcomes**
Students compare normal conditions, high UV, and rapid division scenarios to see how different factors affect mutation frequency and fitness.

R **Revise & Extend**
Students add DNA repair mechanisms or cancer risk to explore how organisms defend against and are harmed by mutations.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Your body copies 3 billion letters of DNA every time a cell divides. What if it makes a typo?"*

Watch For

Analogy: Copy a paragraph by hand 1,000 times. How many errors would you make?

Slide 2: Learning Objectives

3 min

Say: *"Today we model what happens when DNA makes mistakes — and why it matters."*

Watch For

Pre-teach 'mutation' vs. 'mutagen' (the cause vs. the result).

Slide 3: Big Question

3 min

Say: *"Most mutations do nothing. Some cause disease. And rarely, some give an advantage. Let's model it."*

Watch For

Quick write: What do you think causes mutations? List everything you know.

Slide 4: LEVER Framework

2 min

Say: *"We'll build a model connecting UV light and cell division to mutations to fitness."*

Watch For

Preview the chain from external inputs to organism-level effects.

Slide 5: Activity 1: Components

8 min

Say: *"What causes mutations? What do mutations affect? Let's sort our components."*

Watch For

Guide sorting. Discuss why UV and cell division are external inputs.

Slide 6: Activity 2: Connections

8 min

Say: *"Both UV and fast division increase mutations. But the effects on fitness can go either way."*

Watch For

Map relationships. Discuss the +/- nature of protein function changes (can help OR hurt).

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's crank up the UV and watch what happens. Then crank up division rate."*

Watch For

Students predict, then run. Compare how each input drives mutation frequency differently.

Slide 8: Discoveries

5 min

Say: *"Most mutations are duds. A few are dangerous. And very rarely, one helps. That's evolution's fuel."*

Watch For

Connect to cancer risk data. Discuss why sunscreen matters at the molecular level.

Slide 9: STEM Challenge

12 min

Say: *"Design a campaign that uses science — not scare tactics — to protect teens from UV damage."*

Watch For

Teams create evidence-based campaigns. Present and evaluate messaging effectiveness.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** UV Radiation Exposure, Cell Division Rate
- **INTERNAL:** Mutation Frequency, Protein Function Change, Organism Fitness

Relationships

- **UV Radiation Exposure to Mutation Frequency:** POSITIVE (+) — More UV radiation causes more DNA damage (thymine dimers and base modifications), increasing the rate at which mutations occur during DNA replication.
- **Cell Division Rate to Mutation Frequency:** POSITIVE (+) — Faster cell division means more DNA copying events, and each copy is another opportunity for errors. More copies = more mistakes.
- **Mutation Frequency to Protein Function Change:** POSITIVE (+) — More mutations mean more chances that a gene's code is altered, producing a protein with a different shape and potentially different function.
- **Protein Function Change to Organism Fitness:** POSITIVE/NEGATIVE (+/-) — Changed proteins can be harmful (disrupting vital functions), neutral (no significant effect), or rarely beneficial (providing new capabilities). The net effect depends on which proteins are affected and how.
- **Organism Fitness to Cell Division Rate:** POSITIVE (+) — Healthier organisms with higher fitness maintain normal growth patterns and cell division. Reduced fitness can alter growth rates and tissue maintenance.

Key Discoveries

- UV radiation damages DNA by causing chemical changes to nucleotide bases
- Faster cell division means more DNA copying events, which means more chances for errors
- Most mutations are neutral — they don't significantly change protein function
- Harmful mutations are more common than beneficial ones, but rare beneficial mutations drive evolution
- Mutation frequency increases with BOTH UV exposure and cell division rate — the risks are additive

Multiple Choice Post-Assessment Answer Key

1. Your model shows that when UV Radiation Exposure is set to 90%, Mutation Frequency increases dramatically and Organism Fitness declines. Which explanation is best supported by the model?

Correct: B) UV radiation damages DNA bases, leading to more copying errors during cell division, which produces more mutations that alter protein function and reduce fitness

Correct! The model traces a causal chain: UV radiation damages DNA, which increases errors during replication, which increases mutation frequency, which alters protein function, which typically reduces organism fitness because harmful mutations outnumber beneficial ones.

2. The model shows that both UV Radiation Exposure and Cell Division Rate independently increase Mutation Frequency. A student asks why skin cells are at higher cancer risk than brain cells. Which model-based explanation is correct?

Correct: B) Skin cells are exposed to UV radiation AND divide rapidly, meaning both external inputs are high, compounding the mutation frequency

Correct! Skin cells face a double risk: they are directly exposed to UV radiation (a mutagen) AND they divide frequently to replace damaged cells. Both model inputs are elevated, creating a compounded increase in mutation frequency compared to cells like neurons that rarely divide.

3. Your model shows that Protein Function Change has a POSITIVE/NEGATIVE (+/-) relationship with Organism Fitness. What does this dual relationship represent?

Correct: B) Most mutations change protein function in ways that are harmful to the organism, but rare mutations can be beneficial, so the effect can go either direction

Correct! The dual sign reflects that mutations can have different outcomes. Most protein changes disrupt normal function (harmful). Some change non-critical regions (neutral). Rarely, a mutation improves a protein's function in the current environment (beneficial). The net effect depends on which proteins are affected.

4. A student claims that UV radiation causes mutations instantly the moment it hits DNA. Using the model, which response is most accurate?

Correct: B) UV radiation damages DNA bases immediately, but the damage becomes a mutation only when the cell divides and copies the damaged DNA incorrectly

Correct! The model shows that Cell Division Rate is a critical link between DNA damage and actual mutations. UV causes chemical damage (like thymine dimers) immediately, but this damage becomes a permanent mutation only during DNA replication. Without cell division, the damage may be repaired before it becomes heritable.

5. Your model shows that mutations are the raw material for evolution, yet most mutations reduce organism fitness. How do these two facts coexist?

Correct: B) Evolution works at the population level over many generations. Most mutations are harmful to individuals, but the rare beneficial mutation spreads through the population via natural selection, driving adaptation.

Correct! The model operates at the individual level, where most mutations reduce fitness. But evolution operates at the population level over many generations. Across millions of individuals, the rare beneficial mutation provides a survival advantage that natural selection amplifies, gradually changing the population.

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Mutations are always harmful"	Most mutations are neutral — they don't change protein function due to genetic code redundancy. Only mutations that alter critical protein regions tend to be harmful. And rarely, mutations provide beneficial new functions that drive evolution.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"UV radiation causes mutations instantly"	UV radiation damages DNA by creating chemical modifications like thymine dimers. These become mutations only when the cell divides and copies the damaged DNA incorrectly. The damage is immediate, but the mutation happens during the next cell division.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"You can feel mutations happening"	Mutations are molecular-level events — you cannot feel a single base pair change. Even mutations that cause disease often take years or decades to produce symptoms. Cancer from UV exposure typically develops 20-30 years after the initial DNA damage.